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Java problem statement for a Tour app that incorporates concepts like inheritance, abstract classes, interfaces, and exception handling:

Problem Statement: Tour App

You are tasked with developing a Tour app that allows users to book and manage tours to different destinations.

The app should support different types of tours, such as Adventure Tour, City Tour, and Wildlife Tour. Each tour type has specific attributes and behaviors.

Requirements:

1. Implement a class called `Tour` that represents a generic tour. It should have the following attributes:

- `tourId` (integer)
- `tourName` (String)
- `tourDuration` (integer)
- `tourCost` (double)

2. Define an abstract class called `AbstractTour` that extends the `Tour` class. It should add the following attributes:

- `tourGuideName` (String)
- `minimumGroupSize` (integer)

The `AbstractTour` class should have the following methods:

- An abstract method called `calculateTotalCost()` that calculates the total cost of the tour.
- A concrete method called `displayTourDetails()` that displays the details of the tour.

3. Implement three classes that extend the `AbstractTour` class:

- `AdventureTour`: It should have an additional attribute called `activityLevel` (String). Implement appropriate methods and constructors.

- `CityTour`: It should have an additional attribute called `city` (String). Implement appropriate methods and constructors.

- `WildlifeTour`: It should have an additional attribute called `wildlifeType` (String). Implement appropriate methods and constructors.

4. Define an interface called `BookingSystem` with the following methods:

- `bookTour()`: This method should take necessary parameters and book the tour.
- `cancelTour()`: This method should cancel a booked tour.

5. Implement the `BookingSystem` interface in the `AbstractTour` class. Handle the booking and cancellation of tours accordingly.

Use appropriate exception handling to handle any exceptions that may occur during booking or cancellation.

6. Implement a main class called `TourApp` that demonstrates the functionality of the Tour app. It should include the following:

- Create instances of different tour types (`AdventureTour`, `CityTour`, `WildlifeTour`) and test their methods.
- Use try-catch-finally blocks to handle exceptions while booking or canceling tours.

Note:

- You are not allowed to use any pre-defined collection classes (e.g., `ArrayList`, `LinkedList`) for storing tour details or bookings. You should manage the data using appropriate variables and arrays if necessary.

- You may assume that all necessary classes and methods not mentioned in the requirements have already been implemented.

Your task is to implement the classes and methods as per the given requirements and ensure that the Tour app functions correctly, including appropriate exception handling.

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